Monopsony in an Elastic Labor Market

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Does Monopsony Matter?

Labor economists often infer the importance of monopsony by estimating the labor supply elasticity facing firms. (Bachmann & Frings ’17; Barth & Dale-Olsen ’09; Blau & Kahn ’81; Dal Bó, Finan & Rossi ’13; Bodah, Burkett & Lardaro ’03; Booth & Katic ’11; Brochu & Green ’13; Caldwell & Oehlsen ’18; Cho ’18; Currie ’91; Van den Berg & Ridder ’98; Depew & Sørensen ’13; Depew, Norlander & Sørensen ’17; Dobbelaere & Mairesse ’13; Dobbelaere & Vancauteren ’17; Dube, Manning & Naidu ’18; Dube, Jacobs, Naidu & Suri ’18; Dube, Giuliano & Leonard ’19; Dube, Lester & Reich ’16; Fakhfakh & FitzRoy ’06; Falch ’10; Falch ’11; Falch ’17; Fleisher & Wang ’04; Galizzi ’01; Gittings & Schmutte ’16; Hirsch & Jahn ’15; Hirsch, Jahn, Manning & Oberfichtner ’19; Hirsch, Jahn & Schnabel ’18; Hirsch, Schank & Schnabel ’10; Hotchkiss & Quispe-Agnoli ’13; Howes ’05; Campbell ’93; Kline, Petkova, Williams & Zidar ’19; Manning ’03; Mas ’17; Matsudaira ’14; Meitzen ’86; Naidu, Nyarko & Wang ’16; Ogloblin & Brock ’05; Pörtner & Hassairi ’18; Ransom & Sims ’10; Ransom & Oaxaca ’10; Staiger, Spetz & Phibbs ’10; Sulis ’11; Sullivan ’89; Tortarolo & Zarate ’18; Tucker ’17; Vick ’17; Wasylenko ’77; Webber ’15; Webber ’16; Webber ’18.)

- When labor supply is elastic, a firm which pays less than its marginal product would lose many workers.
- So a large labor supply elasticity would imply that monopsony doesn’t matter??

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- When labor supply is elastic, a firm which pays less than its marginal product would lose many workers.
- So a large labor supply elasticity would imply that monopsony doesn’t matter??

Nope:

- When labor supply is elastic, even modest markdowns could substantially change the equilibrium allocation of workers.

This paper: **As labor supply becomes more elastic, monopsony distortions can *grow*.**

Does Monopsony Matter?

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I measure distortions first in terms of *quantities*...

- We can postulate a labor supply system while being agnostic about microfoundations.
- As labor supply becomes less elastic, the monopsonistic distortions disappear.
- As labor supply becomes more elastic, the monopsonistic worker allocation will converge — **but not to the efficient allocation**.

Does Monopsony Matter?

This paper: **As labor supply becomes more elastic, monopsony distortions can *grow*.**

I measure distortions first in terms of *quantities*...

... and then in terms of *deadweight loss*.

- This requires an explicit microfoundation for labor supply schedules.
- But I show that a random utility model and a search model yield **identical expressions for the deadweight loss**.
- **More elastic labor supply can also increase deadweight loss**.

The Perturbative Approach

Monopsony can introduce many different distortions —

- Too much unemployment,
- Too many workers assigned to low-productivity firms,
- Too many workers assigned to *high*-productivity firms! (Volpe ’25)

— for many different reasons —

- Heterogeneous preferences,
- Concentration,
- Search frictions.

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— for many different reasons —

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To make claims in full generality, I will mostly take a perturbative approach —

1. Take a first- or second-order approximation to derive formulae relating distortions to elasticities,
2. Ask how those formulas behave as the elasticities grow.

— but at the end of the talk I will illustrate my results using a fully specified model.

Existing literature

The structural literature measuring **monopsonistic distortions.** (Berger, Herkenhoff & Mongey '22; Chan, Kroft, Mattana & Mourifié '24; Volpe '25; Townsend & Allan '25)

⇒ I analyze the determinants of these distortions qualitatively.

Sufficient statistics for discrete choice models. (Chetty '09; Small & Rosen '81)

⇒ I generalize these results to avoid functional assumptions and encompass more general distortions, and by extending the analysis to search models.

Comparisons of the search and random utility microfoundations for market power (Trottner '25; Menzio '24)

⇒ I show that the **two foundations are equivalent**, conditional on certain statistics.

Quantities

Let's **start with a model of a single firm**, with

- A constant returns to scale production function $y(L) = pL$.
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The **competitive wage** w^C ensures that labor supply is equal to the labor that the firm would demand, were it to take the wage as given:

$$L\left(w^C\right) \in \arg\max_{L \in \mathbb{R}^+} \left\{y(L) - w^C L\right\} \quad \Rightarrow \quad w^C = p.$$

The **monopsony wage** w^M maximizes the firm's profit, given its labor supply constraint:

$$w^M \equiv \arg\max_{w \in \mathbb{R}^+} \left\{y(L(w)) - wL(w)\right\} = p\frac{\eta}{1+\eta},$$

where $\eta \equiv \frac{w}{L} L'(w)$ is the labor supply elasticity facing the firm.

Quantities

Combining our wage expressions, $w^M = w^C \frac{\eta}{1+\eta}$, and so:

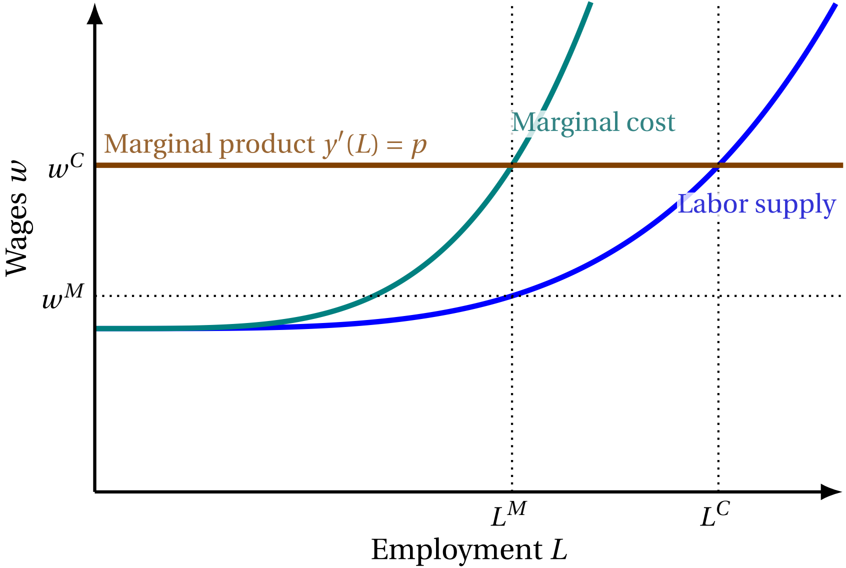
$$w^C - w^M = w^C \left(1 - \frac{\eta}{1+\eta}\right) = \boxed{w^C \left(\frac{1}{1+\eta}\right)}.$$

So to first order, the log **distance between the competitive labor utilization** $L^C = L\left(w^C\right)$ **and the monopsonistic labor utilization** $L^M = L\left(w^M\right)$ is

$$\log L^C - \log L^M \approx \frac{d \log L}{dw}\left(w^C - w^M\right) = \frac{d \log L}{dw} \boxed{w^C \left(\frac{1}{1+\eta}\right)} = \frac{\eta}{1+\eta},$$

which is monotonically increasing in η.

Quantities



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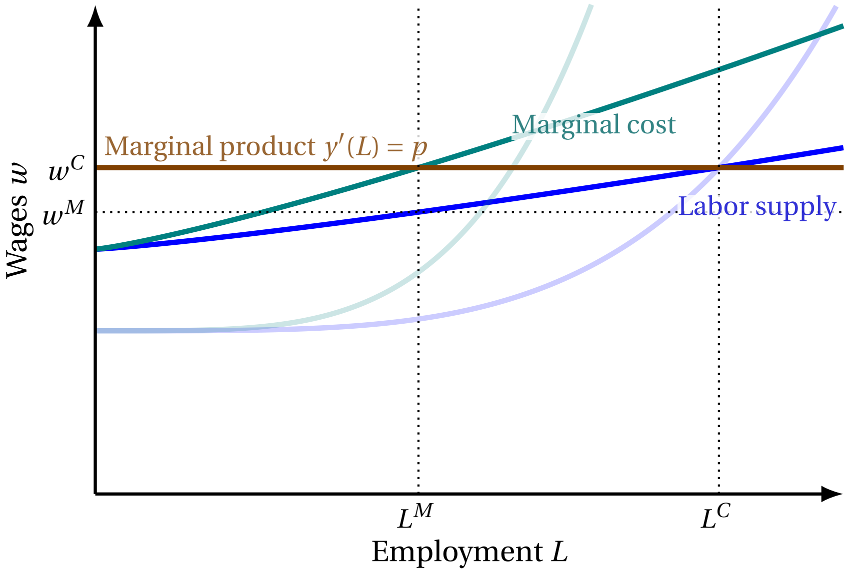
Marginal product $y'(L) = p$
w^C
Wages w
Labor supply
w^M
L^M
L^C
Employment L

Quantities



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The **analysis with many firms** $f \in \mathbf{F}$ is very similar. I assume

- Each firm has a constant returns to scale production function $y_f(L) = p_f L$.
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The **competitive wages** $\vec{w}^C$ ensure that labor supply is equal to the labor that each firm would demand, were it to take the wage as given:

$$L_f\left(w_f^C, \vec{w}_{-f}^C\right) \in \arg\max_{L \in \mathbb{R}^+} \left\{y_f(L) - w_f^C L\right\} \quad \Rightarrow w_f^C = p_f.$$

The **monopsony wages** $\vec{w}^M$ maximize each firm's profit, given the labor supply system:

$$w_f^M \equiv \arg\max_{w \in \mathbb{R}^+} \left\{y_f\left(L_f\left(w; \vec{w}_{-f}^M\right)\right) - w L_f\left(w; \vec{w}_{-f}^M\right)\right\} = p_f \frac{\eta_{f,f}}{1+\eta_{f,f}},$$

where $\eta_{f,f}$ is the firm's own-wage labor supply elasticity.

Quantities

Combining our wage expressions, we have that $w_g^M = w_g^C \frac{\eta_{g,g}}{1+\eta_{g,g}}$, and so:

$$w_g^C - w_g^M = w_g^C \left(1 - \frac{\eta_{g,g}}{1+\eta_{g,g}}\right) = w_g^C \left(\frac{1}{1+\eta_{g,g}}\right).$$

So to first order, the log **distance between the competitive labor allocation** $L_f^C = L_f\left(w_f^C, \vec{w}_{-f}^C\right)$ **and the monopsonistic labor allocation** $L_f^M = L_f\left(w_f^M, \vec{w}_{-f}^M\right)$ is

$$\log\left(L_f^C\right) - \log\left(L_f^M\right) \approx \sum_{g \in \mathbf{F}} \frac{\partial \log L_f}{\partial w_g}\left(w_g^C - w_g^M\right) = \sum_{g \in \mathbf{F}} \frac{\partial \log L_f}{\partial w_g} w_g^C \left(\frac{1}{1+\eta_{g,g}}\right)$$

$$= \sum_{g \in \mathbf{F}} \frac{\eta_{f,g}}{1+\eta_{g,g}},$$

where $\eta_{f,g}$ is the elasticity of firm f's labor supply with respect to firm g's wage.

Quantities

$$\log\left(L_f^C\right) - \log\left(L_f^M\right) \approx \sum_{g \in \mathbf{F}} \frac{\eta_{f,g}}{1+\eta_{g,g}}.$$

How is misallocation affected by the matrix of elasticities?

- Consider a proportionate change to each of the cross-wage and own-wage elasticities: $\eta_{f,g} = \alpha\,\eta^0_{f,g}$.

Quantities

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How is misallocation affected by the matrix of elasticities?

- Consider a proportionate change to each of the cross-wage and own-wage elasticities: $\eta_{f,g} = \alpha\, \eta^0_{f,g}$.
- **As the elasticities become small, the misallocation will approach zero:**

$$\lim_{\alpha\to 0^+}\left[\log\left(L_f^C\right) - \log\left(L_f^M\right)\right] = \lim_{\alpha\to 0^+} \sum_{g\in\mathbf{F}} \frac{\alpha\eta^0_{f,g}}{1+\alpha\eta^0_{g,g}} = 0.$$

- **As the elasticities become large, the misallocation will converge to a constant:**

$$\lim_{\alpha\to \infty}\left[\log\left(L_f^C\right) - \log\left(L_f^M\right)\right] = \lim_{\alpha\to \infty} \sum_{g\in\mathbf{F}} \frac{\alpha\eta^0_{f,g}}{1+\alpha\eta^0_{g,g}} = \sum_{g\in\mathbf{F}} \frac{\eta^0_{f,g}}{\eta^0_{g,g}},$$

  which is generically non-zero.

Costs

We've shown that more elastic labor markets suffer more misallocation.

But does the social cost of that misallocation also increase?

You might think this would depend on how we micro-found our labor supply schedule.
It needn't!

Costs: Random utility

First consider a random utility framework, in which **labor supply is governed by workers' preferences** over firms (and non-employment).

Each worker $i \in [0,1]$ chooses either a firm or nonemployment to maximize their utility:

$$f_i(\vec{w}) \in \arg\max_{g \in \mathbf{F} \cup \{\emptyset\}} \left\{ v\left(w_g\right) + e_{i,g} \right\},$$

- $\emptyset$ denotes nonemployment,
- $v(\cdot)$ is a smooth, increasing function,
- the random vectors $\vec{e}_i \equiv \left(e_{i,f}\right)_{f \in \mathbf{F} \cup \{\emptyset\}}$ are drawn from some absolutely continuous distribution, independent across workers.

These preferences generate the additive random utility model

$$L_f\left(w_f, \vec{w}_{-f}\right) = \mathbb{P}\left[f \in \arg\max_{g \in \mathbf{F} \cup \{\emptyset\}} \left\{ v\left(w_g\right) + e_{i,g} \right\} \right].$$

Costs: Random utility

We measure costs using **compensating variation**.

- Let $f_i^C \equiv f_i\left(\vec{w}^C\right)$ denote worker i's competitive firm.
- Let $\mathrm{CV}_i(\vec{w})$ denote the transfer required to compensate worker i for the wage vector $\vec{w}$, rather than the competitive wage:

$$v\left(w^C_{f_i^C}\right) + e_{i,f_i^C} = \max_{f \in \mathbf{F} \cup \{\emptyset\}} \left\{ v\left(w_f + \mathrm{CV}_i\left(\vec{w}\right)\right) + e_{i,f} \right\}.$$

- Let $\mathrm{CV}_f(\vec{w})$ denote the transfer required to compensate firm f for the wage vector $\vec{w}$, relative to the competitive wage:

$$L_f\left(w_f^C, \vec{w}_{-f}^C\right)\left(p_f - w_f^C\right) = L_f\left(w_f, \vec{w}_{-f}\right)\left(p_f - w_f\right) + \mathrm{CV}_f(\vec{w}).$$

We measure deadweight loss as $\mathcal{C}\left(\vec{w}\right) \equiv \int_{i \in \mathbf{I}} \mathrm{CV}_i \, di + \sum_{f \in \mathbf{F}} \mathrm{CV}_f$.

Costs: Random utility

First consider workers:

$$v\left(w^{C}_{f^C_i}\right)+e_{i,f^C_i}=\max_{f\in\mathbf{F}\cup\{\emptyset\}}\left\{v\left(w_f+\mathrm{CV}_i\left(\vec{w}\right)\right)+e_{i,f}\right\}.$$

Let $f_i^*=\arg\max_{f\in\mathbf{F}\cup\{\emptyset\}}\left\{v\left(w_f+\mathrm{CV}_i\right)+e_{i,f}\right\}$.

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Let $f^*_i=\arg\max_{f\in\mathbf{F}\cup\{\emptyset\}}\left\{v\left(w_f+\mathrm{CV}_i\right)+e_{i,f}\right\}$.

By the Milgrom & Segal envelope theorem, almost everywhere:

$$\frac{\partial \mathrm{CV}_i}{\partial w_g}=\begin{cases}-1 & \text{if } g=f^*_i\\ 0 & \text{otherwise.}\end{cases}$$

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Aggregating across all workers:

$$\boxed{\frac{\partial}{\partial w_g}\int_{i\in\mathbf{I}} \mathrm{CV}_i \, di = \int_{i\in\mathbf{I}} \frac{\partial \mathrm{CV}_i}{\partial w_g} di = -\mathbb{P}\left[g = f_i^*\right] = -L_g.}$$

Costs: Random utility

Next consider a firm.

Competitive profits will be zero, and so CV_f will equal the negative of a firm's profits:

$$\mathrm{CV}_f = L_f\left(w_f, \vec{w}_{-f}\right)\left(w_f - p_f\right).$$

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Differentiating with respect to the wages of firm g yields

$$\frac{\partial \mathrm{CV}_f}{\partial w_g} = \begin{cases} \left(w_f - p_f\right)\frac{\partial L_f}{\partial w_g} & \text{if } f \neq g \\ \left(w_f - p_f\right)\frac{\partial L_f}{\partial w_f} + L_f & \text{if } f = g. \end{cases}$$

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And so firm g's wage will affect aggregate profits by

$$\boxed{\frac{\partial}{\partial w_g}\sum_{f\in\mathbf{F}} \mathrm{CV}_f = \sum_{f\in\mathbf{F}} \frac{\partial \mathrm{CV}_f}{\partial w_g} = \sum_{f\in\mathbf{F}} \left(w_f - p_f\right)\frac{\partial L_f}{\partial w_g} + L_g.}$$

Costs: Random utility

Combining our results for workers and firms:

$$\frac{\partial \mathcal{C}}{\partial w_g} = \sum_{f \in \mathbf{F}} \left(w_f - p_f\right) \frac{\partial L_f}{\partial w_g} + L_g - L_g$$

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$$\frac{\partial \mathcal{C}}{\partial w_g} = \sum_{f \in \mathbf{F}} \left(w_f - p_f\right) \frac{L_f}{w_g} \eta_{f,g} + L_g - L_g$$

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— which is zero to first order around $\vec{w} = \vec{w}^C = \vec{p}$. Taking a second derivative:

$$\frac{\partial^2 \mathcal{C}}{\partial w_g \partial w_k} = \frac{L_k}{w_g} \eta_{k,g} + \sum_{f \in \mathbf{F}} \left(w_f - p_f\right) \frac{\partial}{\partial w_k} \left[\frac{L_f}{w_g} \eta_{f,g} \right].$$

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So to *second* order around $\vec{w} = \vec{w}^C = \vec{p}$, the costs of non-competitive wage-setting are

$$d\mathcal{C} \approx \frac{1}{2} \sum_{g \in \mathbf{F} \cup \{\emptyset\}} \sum_{k \in \mathbf{F} \cup \{\emptyset\}} \boxed{\frac{L_k^C}{p_g} \eta_{k,g}} \left(w_g - w_g^C \right) \left(w_k - w_k^C \right)$$

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In particular, the costs of *monopsonistic* wage-setting will be

$$\boxed{d\mathcal{C} \approx \frac{1}{2} \sum_{g \in \mathbf{F}} \sum_{k \in \mathbf{F}} L_k^C p_k \eta_{k,g} \left(\frac{1}{1 + \eta_{g,g}} \right) \left(\frac{1}{1 + \eta_{k,k}} \right)}$$

Costs: Search

We now consider a search framework similar to Burdett-Mortensen:

- Wages are time-invariant, and chosen to maximize steady-state profits;
- Job offers are exogenous (given a worker's current employment state).

Unlike Burdett-Mortensen:

- The set of firms is discrete rather than continuous.
- Workers face random moving costs.
- I situate the model in discrete time.

Deadweight loss in a search framework

At the start of period t, worker i will be employed ($f_{i,t} \in \mathbf{F}$) or unemployed ($f_{i,t} = \varnothing$). She then:

Receives flow utility $w_{f_{i,t}}$

→

Receives job offers;
Learns whether she is laid off.

Let $A_{i,t}$ denote her choice set of employment options.

→

Chooses a new employer $f_{i,t+1} \in A_{i,t}$.

If $f_{i,t+1} \neq f_{i,t}$, she pays moving cost c_f.

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Choice sets $A_{i,t}$ and moving costs $\vec{c}_{i,t}$ may depend on her current firm, but are otherwise independent of her history.

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Choice sets $A_{i,t}$ and moving costs $\vec{c}_{i,t}$ may depend on her current firm, but are otherwise independent of her history.

So her problem can be modeled with the Bellman

$$V_f = w_f + \beta \mathbb{E}\left[\max_{g \in A}\{V_g - c_g\} \,\middle|\, f_{i,t} = f\right].$$

A representative agent representation of the search framework

Consider the worker's problem, given a choice set A and a vector of moving costs $\vec{c}$.

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Consider the worker's problem, given a choice set A and a vector of moving costs $\vec{c}$.

Deconstrain the problem, with infinite moving costs for options $\notin A$.

$$= \mathbb{E}\left[\max_{g \in \mathbf{F} \cup \{\emptyset\}} \{V_g - \tilde{c}_g\} \,\middle|\, f_{i,t} = f \right]$$

$$\tilde{c}_g = \begin{cases} c_g & \text{if } g \in A \\ \infty & \text{if } g \notin A \end{cases}$$

A representative agent representation of the search framework

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Deconstrain the problem, with infinite moving costs for options $\notin A$.

... which is equivalent to finding the optimal mapping μ from moving costs to a choice.

$$= \sup_{\mu \in \mathcal{M}} \left\{ \mathbb{E}\left[V_{\mu(\tilde{\vec{c}})} - \tilde{c}_{\mu(\tilde{\vec{c}})} \,\middle|\, f_{i,t} = f \right] \right\}$$

$$\mathcal{M} \equiv [0, \infty]^{|\mathbf{F} \cup \{\varnothing\}|} \to \mathbf{F} \cup \{\varnothing\}$$

A representative agent representation of the search framework

Consider the worker's problem, given a choice set A and a vector of moving costs $\vec{c}$.

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$$= \sup_{\vec{T} \in \Delta^{|\mathbf{F}|}} \left\{ \mathbb{E}\left[V_{\mu^*(\vec{\tilde{c}};\vec{T})} - \tilde{c}_{\mu^*(\vec{\tilde{c}};\vec{T})} \,\middle|\, f_{i,t} = f \right] \right\}$$

$$\mu^*\left(\vec{\tilde{c}};\vec{T}\right) \in \operatorname*{argsup}_{\mu \in \mathcal{M}} \left\{ \mathbb{E}\left[V_{\mu(\vec{\tilde{c}})} - \tilde{c}_{\mu(\vec{\tilde{c}})} \,\middle|\, f_{i,t} = f \right] \right.$$

$$\left. \text{s.t. } \left\{ \mathbb{P}\left[\mu\left(\vec{\tilde{c}}\right) = g \right] \right\}_{g \in \mathbf{F} \cup \{\varnothing\}} = \vec{T} \right\}.$$

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Which, using $\mathbb{E}\left[V_{\mu^*(\vec{\tilde{c}};\vec{T})} \,\middle|\, f_{i,t} = f\right] = \vec{T}'\vec{V}$, simplifies as:

$$= \sup_{\vec{T} \in \Delta^{|\mathbf{F}|}} \left\{ \vec{T}'\vec{V} - C_f\left(\vec{T}\right) \right\}$$

$$C_f\left(\vec{T}\right) \equiv \mathbb{E}\left[\tilde{c}_{\mu^*(\vec{\tilde{c}};\vec{T})} \,\middle|\, f_{i,t} = f\right]$$

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In sum, the worker's problem can be expressed equivalently as

$$\mathbb{E}\left[\max_{g\in A}\{V_g - c_g\}\,\middle|\, f_{i,t} = f\right] = \sup_{\vec{T}\in\Delta^{|\mathbf{F}|}}\left\{\vec{T}'\vec{V} - C_f\left(\vec{T}\right)\right\}$$

and so the worker's Bellman equation can be expressed as

$$V_f = w_f + \beta \sup_{\vec{T}\in\Delta^{|\mathbf{F}|}}\left\{\vec{T}'\vec{V} - C_f\left(\vec{T}\right)\right\}.$$

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Summing across firms, overall worker welfare $W = \sum_{f\in\mathbf{F}\cup\{\emptyset\}} L_f V_f$ can be represented by the value function of a representative worker:

$$W\left(\vec{L}\right) = \vec{L}'\vec{w} + \beta\sup_{T\in\mathcal{T}_{\mathbf{F}\cup\{\emptyset\}}}\left\{W\left(T'\vec{L}\right) - C\left(T;\vec{L}\right)\right\}$$

where $\mathcal{T}_{\mathbf{F}\cup\{\emptyset\}}$ is the set of transition matrices and $C\left(T;\vec{L}\right) = \sum_{f\in\mathbf{F}\cup\{\emptyset\}} L_f C_f\left(T_{[f,:]};\vec{L}\right)$ is the overall moving cost associated with the transition matrix T.

Deadweight loss in a search framework

Take the representative worker's Bellman

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and apply the Envelope Theorem:

$$\frac{\partial}{\partial w_f} W\left(\vec{L}\right) = L_f + \beta \frac{\partial}{\partial w_f} W\left(T^*(\vec{L})'\,\vec{L}\right),$$

where $T^*(\vec{L}) \in \operatorname{argsup}_{T \in \mathcal{T}_{\mathbf{F} \cup \{\emptyset\}}} \left\{ W\left(T'\vec{L}\right) - C\left(T; \vec{L}\right) \right\}$.

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where $T^*(\vec{L}) \in \operatorname{argsup}_{T \in \mathcal{T}_{\mathbf{F} \cup \{\emptyset\}}} \{W(T'\vec{L}) - C(T;\vec{L})\}$.

In particular, around a steady-state allocation of labor $\vec{L}$ such that $T^*(\vec{L})'\vec{L} = \vec{L}$,

$$\boxed{\frac{\partial}{\partial w_f} W(\vec{L}) = \frac{L_f}{1-\beta},}$$

which has an annuity value of L_f.

Normative equivalence between search and random utility

We've shown the first-order effect of a wage change on worker welfare is $L_f \, dw_f$.

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Burdett-Mortensen firms are effectively identical to the firms in the random utility model, so the effects on them are identical as well.

$\Longrightarrow$ The second-order approximation

$$d\mathcal{C} \approx \frac{1}{2} \sum_{g \in \mathbf{F}} \sum_{k \in \mathbf{F}} L_k^C p_k \eta_{k,g} \left(\frac{1}{1+\eta_{g,g}} \right) \left(\frac{1}{1+\eta_{k,k}} \right)$$

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holds in **both** the search framework and the random utility framework.

Conditional on the matrix of supply elasticities and the competitive equilibrium, search and random utility are normatively equivalent.

How robust is the equivalence between search and random utility?

I reckon it's kinda robust.

For workers:

- Envelope theorem intuition will often hold, *e.g.* with endogenous search effort.
- However, making 'moving costs' $C\left(T;\vec{L}\right)$ depend on wages (e.g. due to endogenous vacancy creation) will break the equivalence.

For firms:

- Equivalence will hold provided 'marginal product' p_f can be redefined such that the firm's profits can be expressed as

$$\left(p_f - w_f\right) L_f,$$

- *E.g.* p_f may need to be net of capital costs, recruiting costs.
- Diseconomies of scale change the formula but not the equivalence.

Equivalence would break if there are additional externalities *e.g.* due to congestion.

Costs when elasticities become small or large

Consider our formula

$$dC \approx \frac{1}{2} \sum_{g \in \mathbf{F}} \sum_{k \in \mathbf{F}} L_k^C p_k \eta_{k,g} \left(\frac{1}{1+\eta_{g,g}} \right) \left(\frac{1}{1+\eta_{k,k}} \right)$$

and again consider a proportionate change to each of the cross-wage and own-wage elasticities: $\eta_{f,g} = \alpha\, \eta^0_{f,g}$.

- As the elasticities become small, the costs will approach zero:

$$\lim_{\alpha \to 0^+} dC = 0.$$

- As the elasticities become large, the costs will *also* approach zero:

$$\lim_{\alpha \to \infty} dC = 0.$$

Simulations

Our analysis thus far has been perturbative:

1. Take a first- or second-order approximation to derive formulae relating distortions to elasticities,
2. Ask how those formulas behave as the elasticities grow.

This has allowed us to generalize across many different models.

But there's an obvious concern: **elasticities are themselves equilibrium objects**.

So, let's illustrate our results with a fully-specified search model.

Simulations

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- Receives utility w_f while employed at firm f,
- While unemployed, receives a job offer from a random firm each period,
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Employment matches are destroyed at rate $\delta = 0.014$.
The equilibrium is symmetric $\Rightarrow$ no on-the-job search.

Simulations

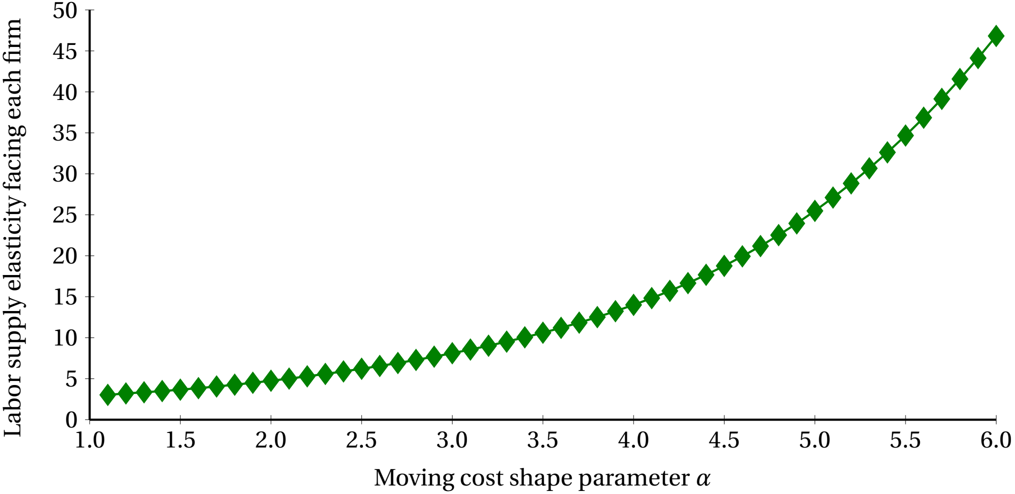
I assume **homogeneous firms**, so the only distortion is **excess unemployment**.

I consider two experiments:

1. Changes to the moving cost shape parameter α.
   - I hold $E[C]$ fixed.
   - This directly (but somewhat artificially) changes the own-wage labor supply elasticity.

2. Reductions to the firms' productivity.
   - Lower productivity $\Rightarrow$ workers are less willing to leave unemployment.
   - This may be a more natural shock to labor supply elasticities.

Simulations: varying shape parameter α



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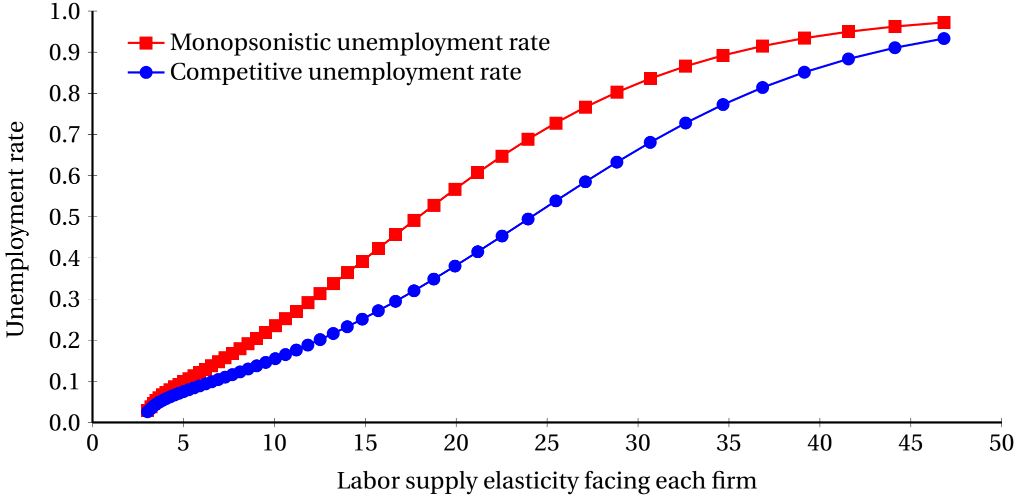

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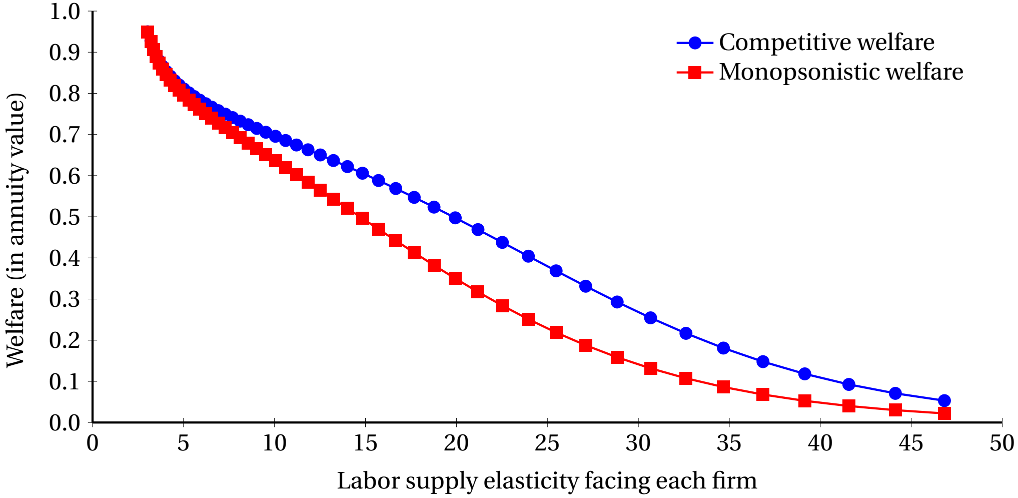

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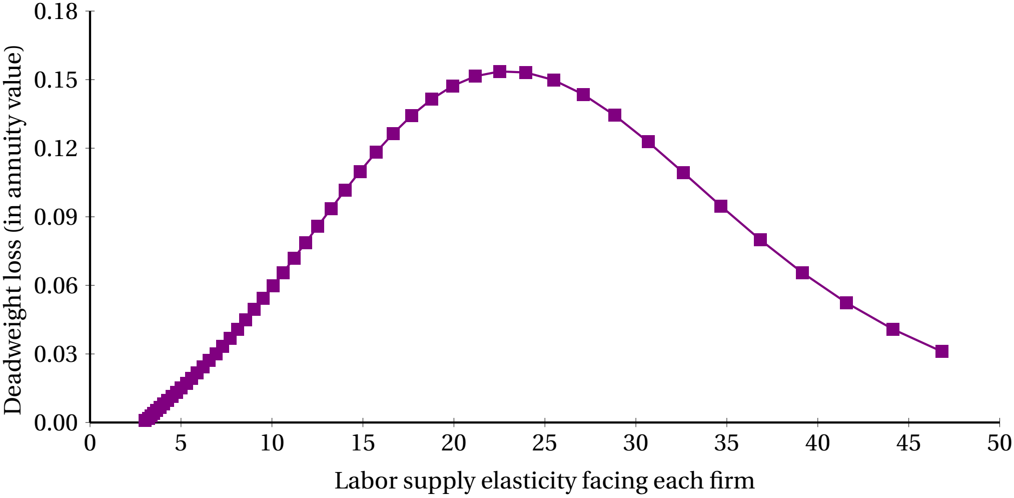
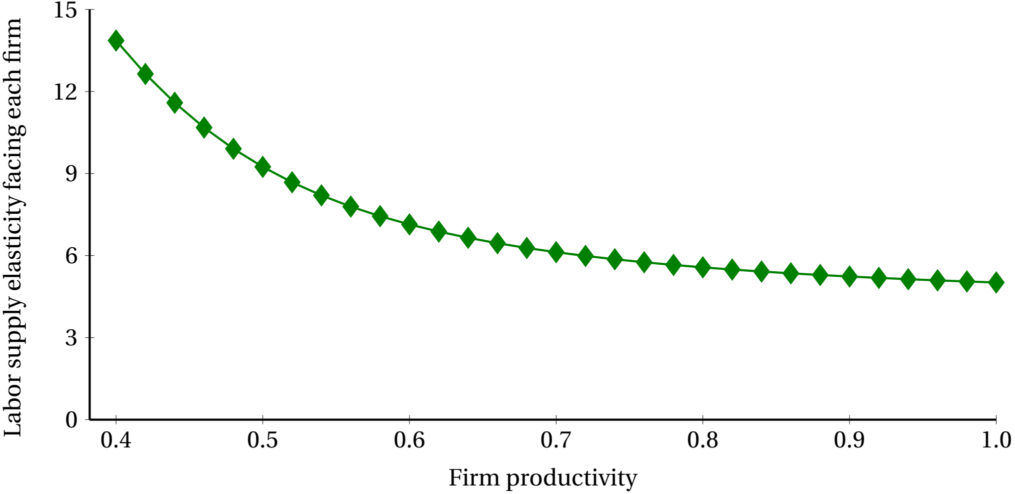


Simulations: varying firm productivity p



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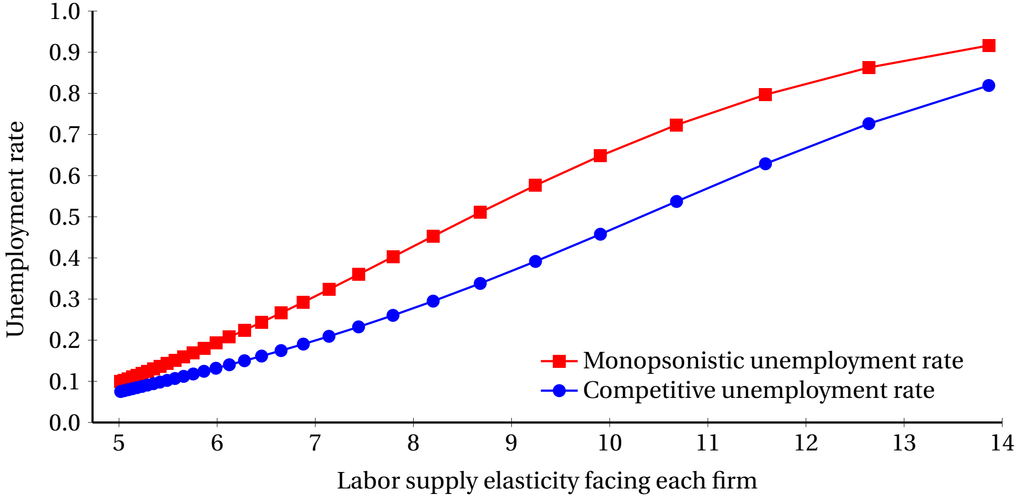

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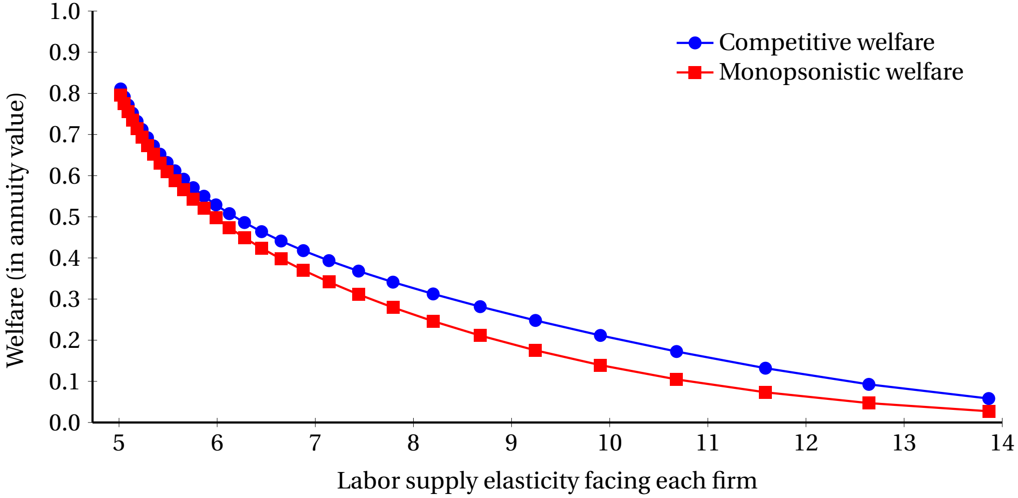

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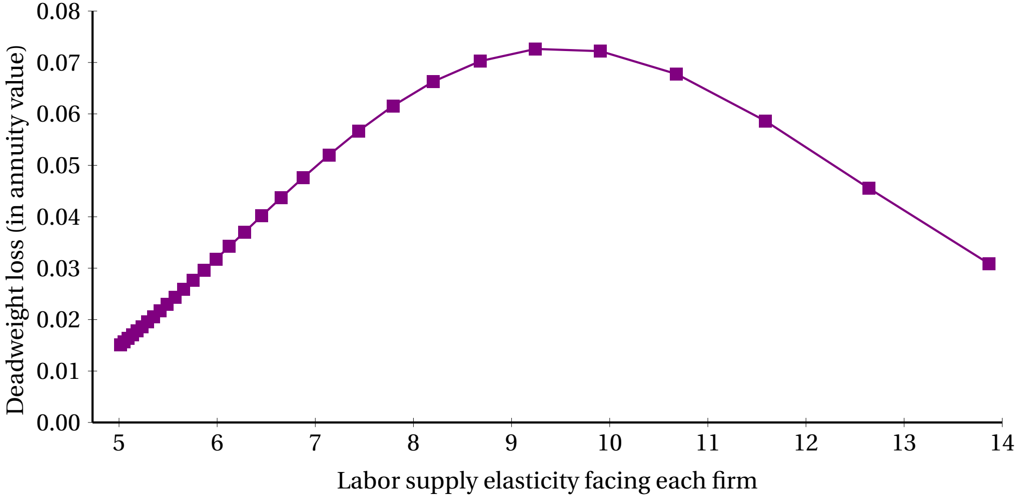

Implications for research

Monopsonistic behavior can have substantial effects, even when firms face large elasticities of labor supply.

- We cannot safely conflate conduct-testing with estimating a structural elasticity.
- *If* we are interested in misallocation, we cannot just look at the own-wage labor supply elasticity.
- Distortions depend on the entire matrix of cross-firm labor supply elasticities.
- Estimation will require *ex ante* restrictions on that matrix.

The welfare consequences of monopsonistic wage-setting are second-order equivalent under both random utility and search.

- But only conditional on the matrix of between-firm labor supply elasticities!
- Rather than asking whether their microfoundations feel natural, we should ask whether they yield the correct elasticities of labor supply.